

Linear Algebra II

Home Assignment V

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1 Home Assignment V (Due: April 19, 2026)

Problem 1.1. (i) Let $\{a_1, a_2, \dots, a_k\}$ be the set of all distinct eigenvalues of a matrix A . Consider the polynomial $q(x) = (x-a_1)(x-a_2) \cdots (x-a_k)$. Show that A is diagonalizable if and only if $q(A) = 0$. (ii) Suppose B is a square matrix and $B^2 = B$. Show that B is diagonalizable.

Solution. We prove the if direction first. Consider a jordan decomposition of A as $J = SAS^{-1}$ having jordan blocks $\{J_{a_i}(n_{i,j})\}_{\substack{1 \leq i \leq k \\ 1 \leq j \leq g_i}}$. With J written in block diagonal form using these jordan blocks, we see that $q(J)$ has $q(J_{a_i}(n_{i,j}))$ as the diagonal blocks. As $q(J) = q(S^{-1}AS) = S^{-1}q(A)S = S^{-1}0S = 0$ we must have that $q(J) = 0$ and thus, $q(J_{a_i}(n_{i,j})) = 0$. It is known that the minimal polynomial of $J_\lambda(m)$ is $(X-\lambda)^m$, so here we must have $(X-a_i)^{n_{i,j}}$ divide $q(X)$ in which the irreducible factor $(X-a_i)$ has multiplicity exactly one forcing $n_{i,j} = 1$. This shows that all the blocks are just $J_{a_i}(1) = [a_i]_{1 \times 1}$ so J is a diagonal matrix and thus A is diagonalisable. The other direction can be shown by first diagonalising A into $D = SAS^{-1}$ with the eigenvalues λ_i on the diagonal and evaluating $q(A) = S^{-1}q(D)S = S^{-1}\text{diag}(q(\lambda_1), \dots, q(\lambda_n))S = S^{-1}0S = 0$ as all the eigenvalues are equal to some a_i here implying $q(\lambda_i) = 0$. $\square$

Problem 1.2. Let q be the minimal polynomial of a complex matrix A . Determine the minimal polynomial of A^* .

Solution. For a polynomial $P(x) = \sum a_n x^n$ define its conjugate as $\overline{P}(x) = \sum \overline{a_n} x^n$. We can easily observe that for a complex matrix A we have $P(A)^* = \overline{P}(A^*)$ (this follows from the common properties of the transpose: skew-linearity, commutativity with powers). Thus, if Q is the minimal polynomial for A and let the one for A^* be P then we find that $\overline{Q}(A^*) = Q(A)^* = 0$ and hence $\overline{Q}|P$ and similarly we see that, $\overline{P}|Q$ so P and Q have the same degree. Now, as $\overline{Q}$ annihilates A^* and has the same degree as P we must have that $P = \overline{Q}$. $\square$

Problem 1.3. Show that the matrix

$$M = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

has no square root, that is, there is no matrix N such that $N^2 = M$. (Hint: $M^2 = 0$.)

Solution. Suppose such N did exist, then we clearly find that $N^4 = 0$ so the minimal polynomial of N divided X^4 and as A is 2×2 , has degree at most 2 and thus its either of X, X^2 and in both cases $N^2 = 0$ which is a contradiction as $M \neq 0$ $\square$

Problem 1.4. Let P be the space of real polynomials of degree at most three. Let $D : P \rightarrow P$ be the linear map defined by $Df = f'$, where f' denotes the derivative of f . Find the spectrum, minimal polynomial, characteristic polynomial of D . Is D diagonalizable?

Solution. Consider a basis $\{1, x, x^2, x^3\}$, then we see that the matrix of D with respect

to this basis is,

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

From basic calculus its clear that $D^4 = 0$ and thus, the spectrum is $\{0\}$. This implies that the characteristic and minimal polynomials are powers of X ; this make the char poly X^4 as the degree of the characteristic polynomial is always equal to the size of the matrix and the minimal polynomial has to be X^4 too as lower powers don't work: $D^3(x^3) = 6 \neq 0$. By [Problem 1.1](#), for it to be diagonalizable we must have that $q(D) = 0$ where $q(X) = X - 0$ but this is clearly not the case so we are done. $\square$

Problem 1.5. Write down the spectral decomposition in the form $\sum_j a_j P_j$, where the P_j 's are mutually orthogonal projections, for the following matrices:

$$A = \begin{pmatrix} 3+i & 1 \\ -i & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 2 & 0 \\ 2 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix}.$$

Solution. To compute the spectral projections via the outer products of orthonormal eigenvectors $P_\lambda = \sum u_i u_i^*$:

For A , the eigenvalues are $\lambda = 4$ and $\lambda = 2$. For $\lambda = 4$, solving $(A - 4I)x = 0$ gives the normalized eigenvector $u_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} i \\ 1 \end{pmatrix}$. The projection is:

$$P_4 = u_1 u_1^* = \frac{1}{\sqrt{2}} \begin{pmatrix} i \\ 1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} -i & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}$$

For $\lambda = 2$, solving $(A - 2I)x = 0$ gives the normalized eigenvector $u_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} -i \\ 1 \end{pmatrix}$. The projection is:

$$P_2 = u_2 u_2^* = \frac{1}{\sqrt{2}} \begin{pmatrix} -i \\ 1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} i & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}$$

The spectral decomposition is $A = 4P_4 + 2P_2$.

For B , the eigenvalues are $\lambda = 1$ and $\lambda = 5$ (multiplicity 2). For $\lambda = 1$, solving $(B - I)x = 0$ gives the normalized eigenvector $v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$. The projection is:

$$P_1 = v_1 v_1^* = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 & 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

For $\lambda = 5$, an orthonormal basis for the eigenspace is $v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $v_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

The projection is the sum of their outer products:

$$P_5 = v_2 v_2^* + v_3 v_3^* = \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

The spectral decomposition is $B = 5P_5 + P_1$. $\square$

Problem 1.6. Let N be a normal matrix. Show that a matrix M commutes with N if and only if it commutes with N^* .

Solution. It was shown in class that for normal N , N^* is a polynomial in N and thus commutativity with N implies that with N^* and we can also go the other way around as $(N^*)^* = N$ to say that commutativity with N^* implies the same with N so we are done. $\square$

Problem 1.7. Write down a matrix whose minimal polynomial is $q(x) = (x - 2)(x - 3)(x + 4)$ and the characteristic polynomial is $p(x) = (x - 2)(x - 3)(x + 4)^2$.

Solution.

$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & -4 & 0 \\ 0 & 0 & 0 & -4 \end{bmatrix}$$

works. $\square$

Problem 1.8. Write down polar and singular value decompositions for the following matrix:

$$C = \begin{pmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 4 & 0 \end{pmatrix}.$$

Solution. First, compute $C^T C$ to find the positive semi-definite matrix $P = \sqrt{C^T C}$:

$$C^T C = \begin{pmatrix} -2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 4 & 0 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 25 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$P = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Next, find the orthogonal matrix $U = [u_1 \ u_2 \ u_3]$ from $C = UP$. Using the non-zero columns of P :

$$2u_1 = \begin{pmatrix} -2 \\ 0 \\ 0 \end{pmatrix} \implies u_1 = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}$$

$$5u_2 = \begin{pmatrix} 0 \\ 3 \\ 4 \end{pmatrix} \implies u_2 = \begin{pmatrix} 0 \\ \frac{3}{5} \\ \frac{4}{5} \end{pmatrix}$$

Choose u_3 to complete the orthonormal basis for $\mathbb{R}^3$:

$$u_3 = \begin{pmatrix} 0 \\ -\frac{4}{5} \\ \frac{3}{5} \end{pmatrix} \implies U = \begin{pmatrix} -1 & 0 & 0 \\ 0 & \frac{3}{5} & -\frac{4}{5} \\ 0 & \frac{4}{5} & \frac{3}{5} \end{pmatrix}$$

This gives the polar decomposition $C = UP$.

To find the SVD, we orthogonally diagonalize P . Since P is already diagonal, use a permutation matrix V to sort the eigenvalues in descending order for Σ :

$$P = V\Sigma V^T = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 5 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Substitute P into the polar decomposition to get $C = U(V\Sigma V^T) = W\Sigma V^T$. Compute the left singular matrix $W = UV$:

$$W = \begin{pmatrix} -1 & 0 & 0 \\ 0 & \frac{3}{5} & -\frac{4}{5} \\ 0 & \frac{4}{5} & \frac{3}{5} \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 & 0 \\ \frac{3}{5} & 0 & -\frac{4}{5} \\ \frac{4}{5} & 0 & \frac{3}{5} \end{pmatrix}$$

The final singular value decomposition $C = W\Sigma V^T$ is:

$$C = \begin{pmatrix} 0 & -1 & 0 \\ \frac{3}{5} & 0 & -\frac{4}{5} \\ \frac{4}{5} & 0 & \frac{3}{5} \end{pmatrix} \begin{pmatrix} 5 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

□

Problem 1.9. Obtain simultaneous diagonalization for the following commuting matrices:

$$E = \begin{pmatrix} 5 & 2 & 0 \\ 2 & 5 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \quad F = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix}.$$

Solution. To construct the orthogonal projections onto the eigenspaces, we find an orthonormal basis $\{x_i\}$ for each eigenspace and compute the sum of their outer products: $P_\lambda = \sum x_i x_i^*$.

For E , the eigenvalues are $\lambda = 7$ and $\lambda = 3$ (multiplicity 2). The normalized eigenvector for $\lambda = 7$ is $x_1 = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0)^T$. Its projection is:

$$P_7 = x_1 x_1^* = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

For $\lambda = 3$, an orthonormal basis for the eigenspace is $x_2 = (\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0)^T$ and $x_3 = (0, 0, 1)^T$. The projection is the sum of their outer products:

$$P_3 = x_2 x_2^* + x_3 x_3^* = \frac{1}{2} \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

Similarly for F , the eigenvalues are $\mu = 2, 0, 4$. Finding their respective normalized eigenvectors y_1, y_2, y_3 and computing $y_i y_i^*$ yields its spectral projections:

$$Q_2 = \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad Q_0 = \frac{1}{2} \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad Q_4 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Since E and F commute, we find the 1D projections onto their joint eigenspaces by multiplying the commuting projections $R_k = P_\lambda Q_\mu$. This isolates the rank-1 outer products $u_k u_k^*$, directly giving us the common orthonormal eigenvectors u_k :

$$\begin{aligned} R_1 = P_7 Q_2 &= \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \implies u_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix} \\ R_2 = P_3 Q_0 &= \frac{1}{2} \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \implies u_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ 0 \end{pmatrix} \\ R_3 = P_3 Q_4 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \implies u_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \end{aligned}$$

Concatenating these basis vectors yields the orthogonal matrix $P = [u_1 \ u_2 \ u_3]$:

$$P = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Applying P simultaneously diagonalizes E and F :

$$P^\top E P = \begin{pmatrix} 7 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \quad P^\top F P = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

□

Problem 1.10. Write E, F of the previous exercise as polynomials of a single matrix G .

Solution. We use the P from the previous problem. Let $A = P D_A P^\top$ where $D_A = \text{diag}(1, 2, 3)$, we will use this as the seed. Notice that for polynomials H we have $H(A) = P H(D_A) P^\top = P \text{diag}(H(1), H(2), H(3)) P^\top$. Thus,

$$A = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{3}{2} & -\frac{1}{2} & 0 \\ -\frac{1}{2} & \frac{3}{2} & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

We use Lagrange interpolation to map the eigenvalues of A to those of E and F . The nodes are $x_1 = 1, x_2 = 2, x_3 = 3$.

For E , the targets are $y_1 = 7, y_2 = 3, y_3 = 3$. The interpolating polynomial is:

$$p_E(x) = 7 \frac{(x-2)(x-3)}{(1-2)(1-3)} + 3 \frac{(x-1)(x-3)}{(2-1)(2-3)} + 3 \frac{(x-1)(x-2)}{(3-1)(3-2)} = 2x^2 - 10x + 15$$

Thus, $E = 2A^2 - 10A + 15I$.

For F , the targets are $y_1 = 2, y_2 = 0, y_3 = 4$. The interpolating polynomial is:

$$p_F(x) = 2 \frac{(x-2)(x-3)}{(1-2)(1-3)} + 0 + 4 \frac{(x-1)(x-2)}{(3-1)(3-2)} = 3x^2 - 11x + 10$$

Thus, $F = 3A^2 - 11A + 10I$.

□

Problem 1.11. (Optional question) State and present a proof of Sylvester's law of inertia. (Hint: See Wikipedia and other sources.)

Solution.

□